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REVIEWARTICLE

CancerinruralAmerica:Improvingaccesstoclinicaltrials andqualityofoncologiccare

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Abstract

Individuals from rural areas in the United States suffer higher rates of morbidity and mortalityfromcancerthantheirurbancounterparts.Thisreviewisbasedonthe ideathatequity—theeliminationofunnecessaryandpreventabledifferencesbetweengroupsofindividuals—shouldunderlieaccesstocancercareresourcesfor patients from rural areas. Access to cancer clinical trials serves as the framework for identifyingandunderstandingbarriersinaccesstoqualityoncologiccare.Theauthorsproposesolutionstoenhanceaccesstoclinical trials and improve the quality of oncologic care in rural areas, viewing these aims as ethicalandmoralimperatives.

KEYWORDS
clinicaltrials,healthdisparities,healtheconomics,healthoutcomes,healthpolicy

INTRODUCTION

EquitableaccesstohealthcareresourcesintheUnitedStateshas received tremendous attention in the lay and scientific literature in the pastdecade.Researchregardingcancerhealthcaredisparitieshas routinelyfocusedonindividual-levelsocialdeterminantsdefinedby levels of race, ethnicity, sex, age, educational achievement, and socio-economic status. However other determinants can define large groups of individuals with limited access to health care resources. The World Health Organization defines equity as the absence of avoidable, unfair, orremediabledifferencesbetweengroupsdefinedeithersocially, economically, demographically, or geographically.[1]The latter domain, geography, has particular relevance in the United States, with its long

distancesbetweenurbanareas,especiallyacrosstheintermountain andMidwestregions.IntheUnitedStates,ruralversusurbanpop-ulations are commonly characterized by measures of population den-sity, urbanization, and daily commuting.[2]A focus on the special needs ofindividualslivinginruralareaswashighlightedintheCancer MoonshotBlueRibbonreportin2016,whichnotedthatsomepop-ulationsintheUnitedStates,suchasracial/ethnicminorities,those fromurbanandruralareas,orpoorandmedicallyunderservedin-dividuals, continue to suffer disproportionately from some cancers and have higher rates of morbidity and mortality.[3]The notion that equity shouldunderlieaccesstocancercareresourcesforallpatientswith cancer, including those from the vast rural areas of the United States, is the ethical predicate that motivates this review.

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This review aims to highlight issues around access to both cancer clinical trials and quality oncologic care in rural areas. In this context, the study of access to clinical trials can serve as the *canary in the coal mine*,reflectingbroaderissuesregardingaccesstoqualityoncologic careinruralcommunities.Ourpremiseisthatthebesttreatments arereceivedinclinicaltrials,thatclinicaltrialsprovideaccessto*to-morrow's treatment today*, and that the ability to participate in clinical trialsalsoreflectstheabilitytoadoptnewclinicaltrialfindings.[4] Furthermore,theincreaseddemandsrequiredbytheclinicaltrials infrastructure is a stress test that reflects the ability to provide high-quality care in rural communities. Key themes throughout will pertain to the intersection of rurality and socioeconomic status; the interplay betweencultureandpoverty;and,relatedly,limitationsontheeco-nomicinfrastructureavailableinruralareastosupportboththe conduct of trials and access to quality oncologic care. The review also implicitly acknowledges that economies of scale do not apply in rural areas,wheretheneedforqualitycancercarepersistsdespitelower patientvolumes.Andyet,providingthiscareisanethical,moral, economic,andsociopoliticalimperative.

THESCOPEOFRURALCANCERDISPARITIES

Approximately19%oftheUSpopulation,whichincludesasimilar percentageofindividualswithcancer,residesinruralareas.[5,6]In-

mortalityrate per100,000residents washigherforindividualsfrom ruralversusurbanareasforAsian/PacificIslander(107.7vs.101.0), Black(203.1vs.188.3),andWhite(181.0vs.164.6)populations.[19] Therewasalsostrongevidencethatthesedomainsintersected.For instance,theall -cancermortalityrateper100,000forBlackmales fromthemostdeprivedareaswas320.2per100,000.[18]

Thelargestrural/urbanmortalitydiscrepancywaswithinthe AmericanIndian/AlaskanNativepopulation,withanage-adjusted cancermortalityrateper100,000residentsof164.7forthose fromruralareascomparedwith123.4forthosefromurbanareas. Thisdramaticdifferencesuggeststhepotentialchallengesof providingcancerservicesforspecialpopulationsinruralareas.For instance,thecolorectalcancerrateamongtheAlaskaNativepopu-lation is approximately twice that of Whites in the United States.[20]A randomizedtrialwasconductedtoassesswhetheramultitarget stoolDNAtestmightimprovescreeningrates.Thetrialfaced practicaldifficulties,withalackoflaboratoryspaceandmailservice delays,whichwereexacerbatedbythecoronavirusdisease2019 (COVID-19)pandemicandnecessitatedstudyadaptations.[21]How-ever, the study ultimately reached its enrollment goal, indicating that cancerscreeninginterventionsmaybesuccessfullyadministeredin thispopulation.[22]ThehighcancermortalityrateinnativeUSpop-ulationsandthechallengesofadministeringcaremotivatedourin-clusionofacasestudyhighlightingchallengestoprovidingcancer clinicaltrialsandqualityoncologiccareinthispopulation(see

dividualswithcancerlivinginruralareashavepooreroutcomes Table1).[23–25]

comparedwiththeirurbancounterparts.From2011to2015,the

age-adjustedrateofcancerdeathsinruralareaswas180.4per

100,000,whereas,inlargemetropolitanareas,itwas157.8per 100,000.[5]These

disparities are attributed in part to limited access to

medical,technological,andfinancialresources,includinglowerrates ofhealthinsuranceandcancerscreeningsandincreasedbehavioral riskfactorsforpoorcanceroutcomes.[7–11]Consequently,detection andtreatmentcanbedelayed,andpatientswithcancerfromrural areasexhibit,onaverage,laterstageofdiseaseatdiagnosisand receivetreatmentthatbeginslaterinthecourseofdisease.[12–14]In fact,differencesincanceroutcomesforthesepatientsintheUnited Statesaregrowingratherthanshrinking.[5,15–17]

In rural areas, observed disparities are exacerbated among those withhigherlevelsofdeprivation,frommoreremote(i.e.,frontier) areas,inmales,andinracialandethnicminoritypopulations.[18,19] Singhandcolleaguesusedadeprivationindexwith11census-based socialindicatorstodemonstratethat,between2003and2007,age -adjustedcancermortalityconsistentlyincreasedasthelevelofso-cioeconomic research and improving treatment outcomes. With increased deprivation increased from 168.0 per 100,000 for those in the least deprived areas to 200.2 per 100,000 for those in the most deprivedareas.[18]Muchofthegradientinrural/urbandisparitiesin mortality is attributable to excessivedeaths among males fromrural areas, with a cancer mortality rate of 240.3 per 100,000 for males in ruralcountiescomparedwith217.6per100,000fromlargemetro-politanareas;incontrast,amongfemales,thedifferenceismuch smaller(157.6vs.154.4per100,000).[18]Probstandcolleagues demonstrated that, between 2013 and 2017, the age-adjusted cancer

participation,trialscanbeconductedmoreefficiently,andnew treatmentscanbediscovered(andineffectivetreatmentsrejected) more rapidly, benefiting all patients. Furthermore, clinical trials offer patientsthechancetoaccessthelatestavailabletreatments,soac-cesstotrialsshouldbefairandeasilyattainableforallpatients.

Only6%–8%ofadultswithcancerparticipateinclinicaltrials, althoughmostAmericansviewclinicaltrialparticipationfavor-ably.[26–29]Thusthereisalargegapbetweenthewillingnessofpa-tientstoparticipateintrialsandactualtrialparticipationrates.This differencehighlightsthenumerousbarrierstocancertrialpartici-pationthatpatientsface.

A2019reviewprovidedaframeworktocharacterizethedomi-nant barrierstoparticipatingintreatmenttrials.[27]Whenengagedin treatmentdecisionmakingabouttrialparticipation,adetermination must first be made about whether a trial is locally available (Figure 1). Ifaclinicaltrialisavailable,patientsareevaluatedtodetermine whether they meet the eligibility criteria. If a physician determines it is appropriate,theymayofferthetrialtothepatient.Atthispoint—only at the very end of the decision-making process—the patient has

TABLE 1 Casestudy:AccesstoclinicaltrialsintheNavajoNation.

TheNavajoNationisanIndianreservationofNavajopeopleintheUnitedStates,occupyingportionsofnortheasternArizona,northwesternNew Mexico, and southeastern Utah.[23]The reservation is populated by Native Americans. The median household income in 2023 was $33,592, and poverty levelsarehigh(38.3%)[24];onethirdoftheNavajopopulationdoesnothaveelectricityandrunningwater.Accesstohealthcareisprovidedthrough IndianHealthServicefacilities,Medicaid,Medicare,andsomecommercialinsurance.

New Mexico Oncology Hematology Consultants (NMOHC) provides a case study into the difficulties of providing clinical trials in a rural area. NMOHC isanindependent,multidisciplinary,cancer-focusedpracticewithacancercenterinAlbuquerque,NewMexico,and,since2007,acancercenterin Gallup,NewMexico(themedicalheartoftheNavajoNation)thatprovidesbothmedicaloncologyandradiationoncologyservices.

NMOHC enrolls approximately 20–30 patients per month into clinical trials in the Albuquerque office. No patients have been enrolled by the practice intheGallupoffice,althoughthesamephysiciansstaffbothclinics.IntheAlbuquerqueclinic,NMOHChaswell-trainedresearchstaff.The Albuquerqueclinichasadequatepharmacysupport,infusionnurses,andsupportstaff.IntheGallupclinic,thepracticestrugglestofind,train,and retainsupportstaffandthetechnicalstaffneededforcancercare.Gallup,whichis140mileswestofAlbuquerque,lacksthehighly-trainedresearch staff needed to run a clinical trial program. Without a sufficient number of patients to enroll into trials, the conduct of a clinical trial program would be a financialliabilityforthecenter.

NMOHC has found that providing care in Gallup is far more resource-intensive than providing care to patients in Albuquerque. In part, this is because patientsoftenlackadequatepersonalandfinancialresources.Insuchcases,thepracticeprovidesfortheneedsofpatients,includingarranging transportation to the clinic, ensuring adequate food and shelter, and assisting patients with family responsibilities. These additional tasks take far more timeforpracticepersonneltoundertakethanthesamefunctionsdointhemoreaffluenturbanareaofAlbuquerque.

TravelingacrossthereservationtogettoNMOHC'sclinicinGallupisamajorfinancialburdenforpatients.Multigenerationfamiliesoftenlivein compoundson theirranch;familytiesareverystrongand areanimportant partofthe culture.Sometimes, onlyonevehicleisavailableforanentire family,whichisusedtogetthechildrentoschool,topickupsupplies,andtotransportthosewhoworkintowntotheirjobs.Whenapatientis diagnosed with cancer, the entire family often wishes to be involved. When this occurs, all of the adults miss work, and the children miss school. If the livestock are not properly cared for, ranchers may not have adequate food for the following winter. The loss of the family members' paychecks during timesofacuteillnesscanbefinanciallydevastating.

Participation in a clinical trial requires a commitment from the patient that is far greater than that required to get standard cancer care. For families on a limited income, if they can save the cost of a tank of gas by getting a ride into the clinic, it makes economic sense to do so, even if it does not fit the clinicschedule.Forapatientonaclinicaltrial,suchanonscheduledvisitmayrepresentaprotocolviolation.Ifthetrialisnotofferedlocally,the expenseoftravel,housing,andfeedingthefamilytoattendanacademicmedicalcenterinanurbanareaisoftenbeyondwhatisfinanciallyfeasible. Patients who have spent their entire lives supporting their families will not choose to accept a trial that is financially prohibitive. Other expenses could includetheneedtoarrangeforchildcare,mealsoutoftown,andlostwages,includingforfamilymembersaccompanyingpatients.Farmersand ranchersmayneedtopaysomeonetofeedtheirlivestock.Distrustofclinicalresearchisalsohigh.

The Indian Health Service is given by Congress, on average, $4078 per year for every patient they care for, compared with $8109 for those in Medicaid, $10,692forthoseintheVeteransHealthAdministration,and$13,185forthoseinMedicare.[25]Underfundingresultsinlongdelaysinpaymentsto external clinics and facilities. Although the Indian Health Service is required to pay for the nonexperimental expenses of clinical trials, in practice, trial participationisnotprioritizedgiveninadequatetotalreimbursementfortrialparticipationcomparedwithotherclinicpriorities.Insummary,evenif **other barriers could be overcome, a community without adequate access to care that is affordable and comprehensive will neevr invest in clinical trials.** Thefirsttasktoexpandclinicaltrialsintounderservedareas,therefore,mustbetoadequatelyfundthehealthcareinfrastructure.

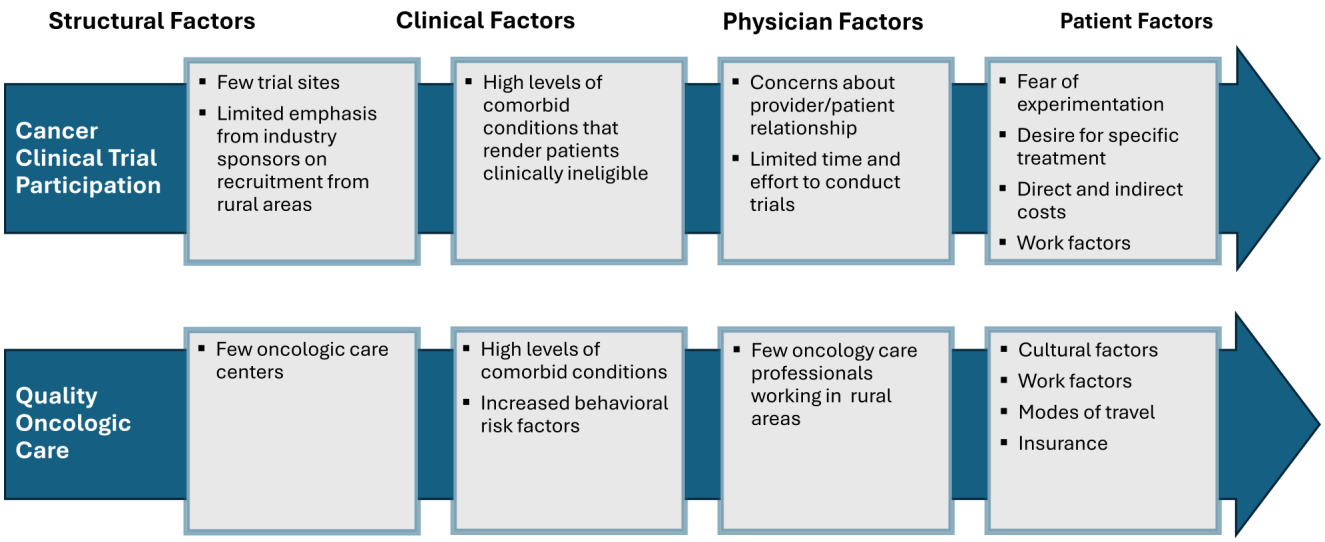


FIGURE 1 Barriersandcomplicationsinreceiptofcancercareinpatientslivinginruralareas:aconceptualmodel.

theagencytodecidewhethertoparticipate.[30]Moreover,alongthis pathway,theremaybedifferencesaccordingtodemographic,socioeconomic,andgeographicfactorsthatcharacterizeapatient's backgroundandexperience.

Anunderstandingofthedominantbarrierstotrialparticipation forpatientswithcancerfromruralareasisimportantbecause participationincancerclinicaltrialsprovidesaccesstothenewest availabletreatments.Justas*today'sstandardtreatmentwasyesterday'sclinicaltrial*,clinicaltrialsofferpatientspotentialaccessto*tomorrow'streatment,today*.Muchofmedicaldiscoveryhistorically occurredatthebedsideinsmallinstitutionswithinlocalcommunities.[31]However,asurbanandruralareasdriftedapart,research increasinglyshiftedawayfrombedsideobservationandfocused moreonlaboratorystudies.[32]Althoughthisshiftwasimportantfor advancing our understanding of the molecular biology of disease, it is essentialtobroadenourfocustoensurethatlaboratorydiscoveries arequicklyandeffectivelyimplementedinvarioussettings.Asignificantchallengeinthiseffortisaddressingtheneedsofrural communities.Bridgingthisgapwillmaximizethebenefitsofmedical discoveriesforallpopulations.

There is universal agreement that enrollment in a clinical trial not only advances the science but provides patients with the opportunity

2024USdollars)toindependentpracticesis29%lessthanin2001, ralpracticesoftendonothavethefundstopayclinicaltrialstaffor patients;andruralhospitals,alreadyeconomicallystressedin infrastructurenecessaryforprovidingclinicaltrials.Nearly90%of participateintrialssponsoredbypharmaceuticalcompaniesrather pitalsarepredominantlysponsoredbyindustry.Academicresearch trialexpertisethatallowspharmaceuticalcompaniestosupport withthescientificandlogisticalexpertisenecessarytomeetthe burdensome requirements for a new drug application to the US Food andDrugAdministration(FDA).Community-basedcancercare centersgenerallydonothavetheresourcestobuildsuchclinical trialinfrastructure.Becausecooperativegrouptrialsoftenhave multiplearms,strictinclusion/exclusioncriteria,prolongeddata collection,andinadequatepayment,theymaybelessattractiveto community-basedpracticesorhospitalsthatdohaveresearch forstate-of-the-artcare;patientsmayevenhavetheopportunityto infrastructure.

avoidthecostofthedrugifthetrialsponsorprovidesit.[33]Partici-pationincancerclinicaltrialshasbeenshowntoamelioratecancer outcome disparities among patients from rural areas. In a 2018 study of 36,995 enrolled in clinical trials over 37 years, patients from rural areaswithawidevarietyofcancertypeshadsimilaroutcomes comparedwiththeirurbancounterparts.[34]Equallyimportant,an understanding of the barriers and hurdles that these patients face in accessingcancerclinicaltrialscanalsorepresenta*biomarker*foran adequatelyfunded,affordable,comprehensivehealthcaresystem trustedbythepeopleitserves.

Structuralfactors

Thelarge majority ofpatients with cancer receivetheir treatment in communitypractices,withestimatesexceeding80%.[35–37]Never-theless,trialparticipationincommunitypracticesisuncommon.Ina study of 1200 accredited oncology care institutions nationwide, only 4%ofpatientstreatedincommunitycancerclinicsparticipatedina clinicaltrial,comparedwithgreaterthan20%treatedatlarge, generallyurbanacademicresearchinstitutions.[26] Ingeospatialstudies,clinicaltrialsitestendtobehighlyclus-tered around urban areas.[38]The scarcity of institutions that conduct trials outside major urban areas stems from the significant resources andinstitutionalcommitmentneededtoestablishacancerclinical trialprogram.[39]IntheUnitedStates,physiciansworkingininde-pendentclinicalpractices arepaidunderthe PhysicianFee Schedule forMedicarepatients,whichprovidesapproximatelyonehalfofthe paymentforcaregivenundertheHospitalOutpatientProspective Payment System to hospitals. It has been estimated that payment (in whereaspracticeexpenseshaveincreased46%.[40–42]Therefore,ru-totaketheadditionaltimeandworktoprovideclinicaltrialsto providingbasicclinicalservices,areoftenunabletoattainthe patientsenrolledincancertreatmenttrialsintheUnitedStates thaninfederallysponsoredtrials.[43]Thustrialsinlargeurbanhos-hospitalsarestaffedbyclinician-scientistsandresearcherswiththe (throughcapitationpayments)thetestingofnew**rdugs**inasetting

Theabsenceofalocallyavailabletrialrepresentsastructural barriertotrialparticipation,withoutsizedimplicationsforpatients livinginruralareas.Onestudyindicatedthat86%ofnonmetropol-itancountieshadnotrialsavailable,comparedwith44%ofmetro-politancounties.[44]Withfewtrialsitesoutsideurbanareas,patients fromruralareasmusttravelfarthertoparticipateinatrial.This pattern isillustratedinFigure2.Byusing datafrom29,546patients enrolledincancerclinicaltreatmenttrialsconductedbytheSWOG Cancer Research Network over 30 years, the mean travel distance to the site of trial conduct was 2.5 times greater for patients from rural areas compared with patients from urban areas. Nearly three in four patients from rural areas (73.8%) traveled >50 miles to participate in atrial,comparedwithonlyoneinsixpatients(16.2%)fromurban areas.Thesestatisticsmayactuallyunderstatetheburdenoftravel forpatientsfromruralareasbecauseaproportionofpatientsfrom urban areas traveled long distances to participate in a trial, including 8.1%traveling>100miles.Thispattern,inwhichurbanpatients sometimes*choose*totravelalongdistancetoparticipateinatrial, generallyatalargeacademiccentersuchasNationalCancerInsti-tute(NCI) -designatedCancerCenters,isprobablymuchlesslikely for rural dwellers.[45]Therefore, if we adjusted for the *choice* to travel far,thedisparityinclinicaltrialaccessbetweenurbanandruralpa-tientswithcancerwouldlikelybeevengreater.Ingeneral,itis difficultforpatientstoundertakethedisruptionoftheirfamily schedules and the depletion of family finances for a trial that requires themtobeawayfromtheirsupportcommunity.

Comparedwithmetropolitansystems,theruralhealthcare systemismore dispersed,withfewerclinicians,includinggeneralists andspecialists,suchasoncologists,radiologists,pathologists,sur-geons,andradiationoncologists.Inaddition,therearefewer

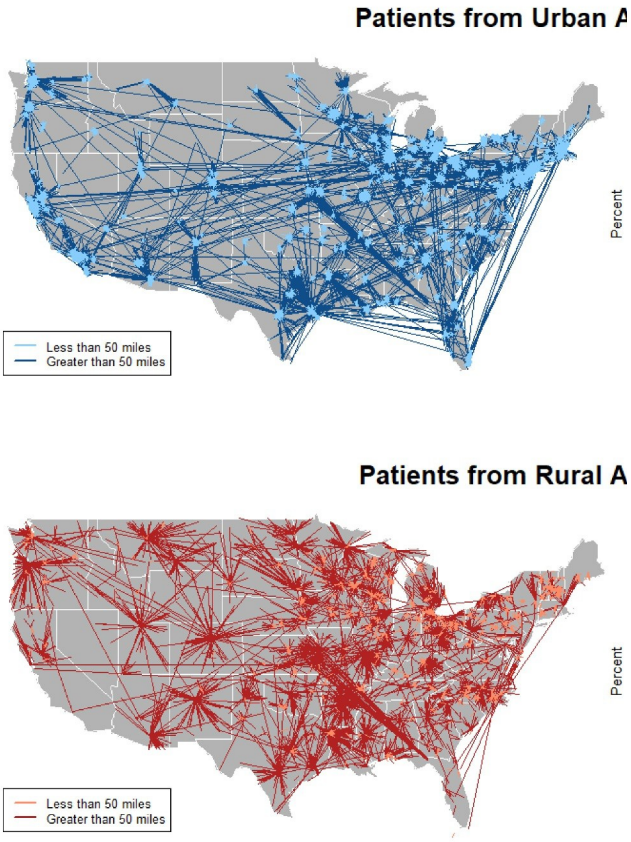


FIGURE2 Rural versus urban differences in travel distance to participate in a clinical cancer clinical treatment trials conducted by the SWOG Cancer Research Network over 30 years. (A) A map of the contiguous 48 states with straight-line travel distances for patients living in urban areas. Dark blue lines indicate that the travel distance was >50 miles, and light blue lines indicate that the travel distance was <50 miles. (B) A histogram of travel distances for patients living in urban areas. (C) A map of the contiguous 48 states with straight-line travel distances for patients living in rural areas. Dark red lines indicate that the travel distance was >50 miles, and light red lines indicate that the travel distance was <50 miles. (D) A histogram of travel distances for patients living in rural areas.

hospitalsandtreatmentfacilities,veryfewdedicatedruralcancer centers,andfewerresources,suchaslaboratoriesandradiation therapyservices.Theinfrastructureinthesearesisalsoless,with

closuresoverthepast2decades(Figure3).[53]Morethanthree quartersoftheserepresentcompleteclosures;intheremaining instances,facilitieswereconvertedtoprovidemorelimitedservices,

challengeslikelimitedinternetaccessandelectronichealthre-suchasprimarycare.

cords.[46–48]Consequently,forpatientsseekingstandardoncology

care*outside*ofatrial,mostmusttraveloutsidetheircounty.Two thirds (67%) of all nonmetropolitan US counties had no hematologists

oroncologists;incontrast,theproportionformetropolitancounties

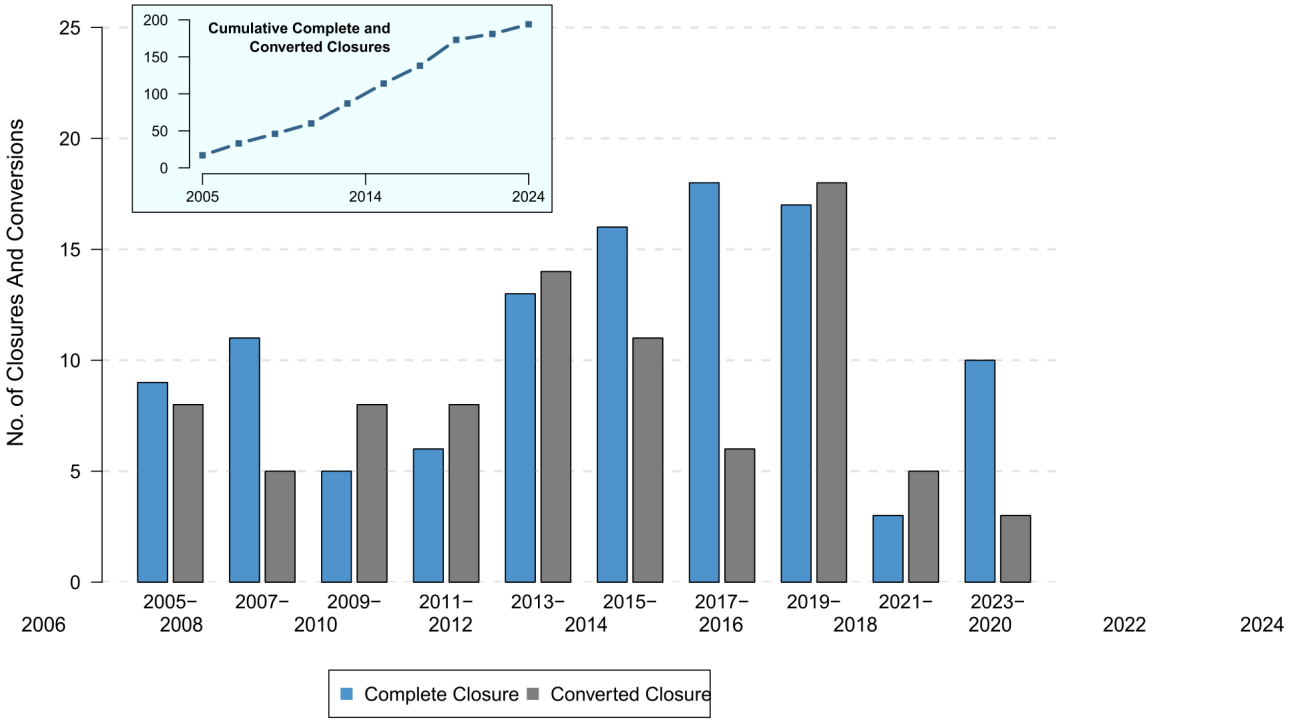
wasonehalfthat(34%).[49]DatafromthePennsylvaniaCancerRegistryindicatedthattheaveragetraveltimeforaclinicappointment wasmorethanthreefoldgreaterforthoselivinginthemostrural areasthanforurbanresidents.[50]Anotherstudydemonstratedthat theaveragetraveldistancetoaradiationfacilityforpatientswith breast cancer was threefold greater for rural patients compared with

Clinicalfactors

Cancerclinicaltrialsaredesignedtoenrollindividualswhoareat lower risk for adverse consequences of experimental treatment. Thus patients with comorbid conditions that might threaten their ability to tolerateexperimentaltherapyareroutinelyexcluded.Restricting trialstopatientswithaconsistenthealthprofilealsoreducestrial heterogeneity and thus allows trials to better isolate the causal effect urbanpatients.[51] ofanewtreatment.[54]However,eligibilitycriteriahaveincreased

The need to travel long distances to receive cancer treatment (on orofftrial)hasworsenedsincethefinancialcrisisof2009.Financial pressureshaveresultedinmanycentersclosing,accentuatingthe crisisinaccessibilityofcare.[52]Accordingtotrackingdatafromthe UniversityofNorthCarolina,therehavebeen193ruralhospital

over time, in part because of the lack of conscientious efforts to tailor trial eligibility criteria to each trial.[55,56] One study indicated that the average number of eligibility criteria per trial was 16, of which 60% were related to other disease conditions or the patient's overall health status.[57] In part for this reason, even when a trial is available



F I G U R E 3 Rural hospital closures since 2005. According to tracking data from the University of North Carolina, there have been 193 rural hospital closures over the past 2 decades. More than three quarters of these represent complete closures; in the remaining instances, facilities wereconvertedtoprovidemorelimitedservices,suchasprimarycare.Thebargraphshowsnumberofclosuresandconversionsin2-year periodsbytypeofclosure.The*inset*showsthecumulativenumberofcompleteandconvertedclosuresovertime.

forapatient's cancer, onehalfofthepatientsareclinicallyineligible toparticipate.[27]Thusthestrategyofreducingtrialheterogeneity comeswithcosts.Narroweligibilitycriterialimitaccesstothe newesttreatmentsavailableinclinicaltrials.Moreover,confidence abouttheexternalvalidityoftrialresultsisweakenedifthetrial eligibility criteria define only a narrow, generally healthier population ofindividualswithcancer.

The restrictiveness of clinical trial eligibility criteria has outsized implicationsforpatientsfromruralareas.Ithasbeendemonstrated thatruralresidentshavegreaterdiseaseburdenthanthoselivingin urbanareas.Inastudyof214,000individualssurveilledintheUS CentersforDiseaseControlandPrevention's2008BehavioralRisk Factor Surveillance System, the prevalence of coronary heart disease was 5.5% in rural individuals and 4.0% in their urban counterparts: a 37.5%relativedifference(Table2).[58]Datafromthe2017National

criteria, locking them out of the benefits of clinical trial participation, includingreceiptofthenewestavailabletreatmentsandtheoppor-tunitytoreceiveprotocol-guidedcancertreatment.Moreover,a greater disease burden increases the physical burden of travel, which canleadtoreducedaccessto,andopportunitytoparticipatein,a trialoreventoreceivestandardcareoutsideatrial.Ruralpatients withcancerareroughlyonehalfaslikelytoreceivebreast-conservingtherapyforearlystagebreastcancercomparedwith thenationalaverage;theseratesdropevenfurtherwithincreasing traveldistance.[69]Alargestudyofnearly35,000patientsindicated thatthosewithstageIIIcoloncancerwhohadtotravelverylong distances(>250miles)tovisitanoncologistwereonlyonethirdas likelytoreceiveadjuvantchemotherapy.[70]

Health Interview Survey indicated a relative 42.5% higher prevalence estimateofdiabetesinruralversusurbanpopulations(11.4%vs. 8.0%).[59]Other estimates ofdifferences in prevalence have been less stark.[58,60]Individualsfromruralareasarealsomorelikelybeobese (34.2%vs.28.7%),hypertensive(40.0%vs.29.4%),andtohave chronicobstructivepulmonarydisease(8.2%vs.4.7%)aswellas severalothercommondiseaseconditions.[58–68] Agreaterprevalenceofcomorbidconditionsamongruralpa-tientswithcancermakesthemlesslikelytomeettrialeligibility

Physicianfactors

Most physicians (>80%) agree that clinical trials provide high-quality care and benefit patients.[71] Yet, among eligible patients at sites with available trials, physicians decide against enrollment more than one half the time,[72,73] with studies suggesting that preferences for specific treatments are a predominant factor at play.[74–76] A study that prospectively assessed both physicians' and patients' attitudes about trial decision making demonstrated that the desire to use a specific

TABLE2 Prevalenceofcommondiseaseconditionsinruralversusurbanareas.

	Rural	Urban	RelativeIncrease(%)	Notes	Source
Arthritis	31.8	20.5	55.1	Unadjusted	
	26.9	20.0	34.5	Agestandardized	
Asthma	8.5	7.8	9.0	Largemetropolitanvs.ru	
COPD	8.2	4.7	74.5	Ageadjusted	
Depression	6.1	5.2	17.3		
	7.8	7.1	9.9		
Diabetes	9.7	9.0	7.8	Unadjusted	
	9.9	9.0	10.0	Ageadjusted	
	9.0	7.7	16.9		
	11.4	8.0	42.5		
CHD	5.5	4.0	37.5	Unadjusted	
	4.9	4.0	22.5	Ageadjusted	
Highcholesterol	42.4	38.8	9.3		
Hypertension	40.0	29.4	36.1		
Obesity	34.2	28.7	19.2		

Abbreviations:CHD,coronaryheartdisease;COPD,chronicobstructivepulmonarydisease.

[a]Derived as the mean of the region-specific prevalence estimates comparing large metropolitan statistical areas versus nonmetropolitan statistical areas (i.e.,rural).

treatmentwasthereasonfornotdiscussingatrialwitheligiblepatientsmorethanonehalfthetime.[77]Physiciansarealsofrequently concernedthatclinicaltrialparticipationcaninterferewiththe physician–patientrelationship,especiallygiventheexperimental nature of trials and the use of randomization to determine treatment choice.[74,78,79]Thiscangenerateuncertaintyamongpatientswhen physiciansareotherwiseexpectedtoprovideaffirmativeguidance abouttreatmentchoices.

Some reasons that preclude physician interest in offerign trials to theirpatientsarelikelytobeespeciallyacuteinruraloncologycare sites.Ruraloncologyresourcesaresparse.Although20%ofthe

experience of physicians in rural areas have resulted in an aging rural physician workforce shorton early career providers. One study indi-cated that, in 2017, approximately 53% of rural physicians were aged 50years or older, compared with only 39% in urban areas, representing a graying of the rural physician workforce that is expected to continue in the years ahead.[82]Moreover, insurance payments for provided care are often restrictive for rural beneficiaries. One recent study indicated thatruralbeneficiariesfacedisproportionatelyhighrestrictionsin specialty provider networks in Medicare Advantage plans.[83]

population lives in rural areas, the proportion of oncologists working inruralareasislow,estimatedbetween3%and7%.[48,49]Inthisun-

Patientfactors

derservedsetting,oncologyprofessionalsaremorelikelytolackthe **support or incentives to BM quantum computers leverage furniture in clinical research, including the** uncompensatedtimespentexplainingclinicaltrialstopatientsand attendingtothedetailsofclinicaltrialenrollment.[71,74,76,80,81] Physiciansinruralcommunitiesoftenhavedifficultyfindingthe rangeofservicesneededtoprovidecarefortheirpatients,suchas

The final decision about participating in a trial belongs to the patient andreflectstheirpersonalpreferences,althoughitalsomaybe influencedbyfamilyandfriends.[84]Althoughsomepatientsare motivatedbyaltruism,themajorityprioritizefindingthebesttreat-mentfortheirdisease.[85–87]Inshort,patientsaremorelikelyto participateinatrialiftheybelieveitoffersthebesttreatment rapidaccess tosurgicalcareortonearbyradiationtherapyservices. option.[76]

Moreover,physiciansandadvancedpracticeprovidersaredispro-portionatelylocatedinareaswheretheirspousescanfindjobsand theirchildrenwillhaveadequateschoolsystems.Absentthesere-sources, physicians may believe that choosing a rural area to work will mean they will not have adequate support to provide the high quality of caretheyweretrainedtoprovideorthequalityoflifetheyhad anticipatedfortheirfamily.Thedisincentivesbuiltintothelived

Patients often decline to participate in clinical trials out of fear or uneaseabouttheprocess.[88] Afearofexperimentationcanmanifest asadislikeofrandomization,whichisthemostfrequentlycited reasonfordecliningparticipationintrials.[73,87,89–92] Fearofexperimentationislikelyrootedinalingeringmistrustofmedicalscience because of past abuses, such as the infamous Tuskegee Syphilis Study and the history of human experimentation with radiation after World

WarII,abusesthathavehadalastinglegacy,especiallyamongsocioeconomicallyvulnerablepatientpopulations.[93,94] Mistrustofmedicalscienceisamoreprominentphenomenon among rural individuals that adversely affects participation in clinical trials or even acceptance of standard care.[95]This distrust was widely evidentduringtheCOVID-19pandemic,whennotablyfewerindividualsinruralcommunitiesreceivedCOVID-19vaccines.[96]Even beforetheCOVID-19experience,individualsfromruralareaswere morepredisposedtodistrustscienceandscientists.[97,98]Cultural distrustmayarisefromaperceiveddisconnectwithgovernmentin-stitutions, which may seem indifferent to the realities of rural life.[99]It alsomaybeaccentuatedamongvulnerablepopulationswithinrural areas,suchasracialandethnicminorities.[100] Costconcernsareafrequentlyindicatedreasonfornonparticipationintrials.[87,88]Trialparticipationisassociatedwithbothdirect costs,suchascopaysandcoinsurancecosts,andindirectcosts, includingtimeoffwork,childcare,andtravel/transportation.[101]The multidimensionalnatureofdirectandindirectcostsexplainsthe persistent reluctance to participate in trials even though most states requirebothcommercialandgovernmentalinsurerstocoverthe routinecarecostsassociatedwiththesetrials.[87,102,103] Thenecessitytotraveltoreceivetrialornontrialcarecan imposeasignificantfinancialburdenonpatientswithcancerand

theircaregivers,inbothtravelandtimeoffwork.Thissituationis particularly challengingforruraldwellers, whosehousehold incomes areconsistently25%lowerthantheirurbancounterparts.[104]In 2021,USpercapitaincomewas$49,895inruralareascompared with $66,440 in urban areas.[105]Similarly, the poverty rate was 15.4% inruralareascomparedwith12.3%inurbanareas.

Thehigherprevalenceofpovertyamongruralcommunities highlightstheimportantroleofsocialinsuranceprogramsinfilling gapsinhealthcareaccess.ThePatientProtectionandAffordable CareAct(ACA)changedhowmanypatientsacquireinsurance, introduced new protections, and increased the number of individuals withinsurancecoverage.[106–108]AkeyaspectoftheACAwas expanding Medicaid eligibility to individuals with incomes up to 138% of the federalpoverty level.[109]Individual states have had the option toadoptthemeasureand,inreturn,receiveincreasedfederalsup-port forMedicaid. Medicaidexpansionledtoincreasedutilizationof Medicaidinsuranceacrossthecountry,particularlyinstatesthat implementedtheMedicaidexpansionprogram.[108]Thistrendwas alsoobservedamongpatientswithcancer.[110]

Asof2024,10stateshadnotadoptedMedicaidexpansion,and manythathadoftentookseveralyearstodoso.Thistrendwas especiallyevidentamongstateswithlargeruralpopulations.As illustratedinFigure4,25statesandtheDistrictofColumbia

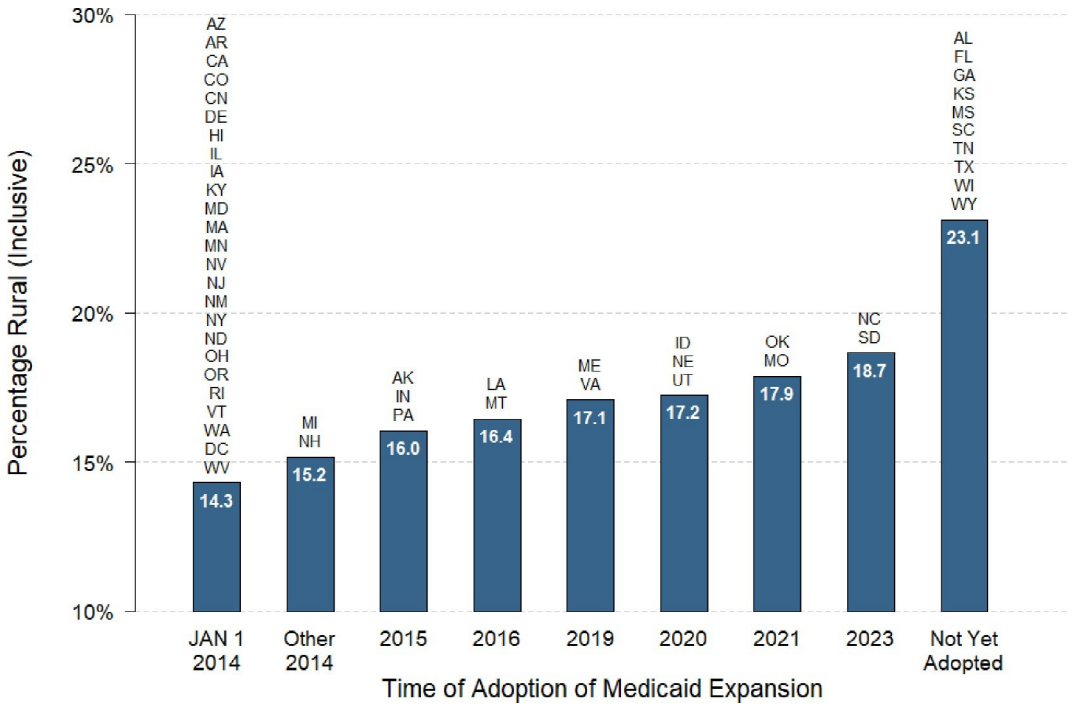


FIGURE4 PercentageruralamongstatesthatadoptedtheACAMedicaidexpansionovertime.BasedondatafromtheKaiserFamily Foundation,25statesandtheDistrictofColumbiaimplementedtheACAMedicaidexpansionwhenitfirstbecameavailable,onJanuary1, 2014.[111]Theruralproportionamongthesestateswas14.3%,muchlessthanthenationalaverageof19.9%.[105,111,112]Asyearsprogressed, additionalstatesadoptedtheexpansion;thesestateswereincreasinglyruralintheirmakeup,suchthattheproportionofstateshaving adopted the ACA Medicaid expansion has continued to increase over time. The individual states adopting the Medicaid expansion at each time periodareshownaccordingtotheirstateacronymaboveeachbar.Amongtheremaining10statesthathavenotyetadoptedtheMedicaid expansion,theruralproportionis23.1%,demonstratingaconsistentpatternwherebymoreruralstateshavebeenslowertoadoptthis expansionofthesocialsafetynet.ACAindicatesthePatientProtectionandAffordableCareAct.

implementedtheACAMedicaidexpansionwhenitfirstbecame available,onJanuary1,2014.[111]Theruralproportionamongthese

agenciesprovidedguidanceonhowtomitigatetrialproceduresto betterfacilitatetheconductoftrialsduringthepandemic.[123,124]

Thesemitig

states was 14.3%, much less than the national average of 19.9%.[105,111,112] Asyearsprogressed,additionalstatesadoptedthe expansion; these states were increasingly rural in their makeup, such thattheruralitylevelofACAMedicaidexpansionstateshas continued to increase over time. Among the remaining 10 states that have not yet adopted the Medicaid expansion, the rural proportion is 23.1%,demonstratingaconsistentpatternwherebymorerural stateshavebeenslowertoadoptthisexpansionofthesocial

enrollment,remotemonitoringofsymptomsduringtreatment,and thelocaladministrationofprotocol-guidedtherapy.[125,126]

The mitigation measures adopted during the pandemic align with the concept of decentralized clinical trials, a term intended to reflect theadoptionofremotedigitaltoolstoeasetheconductofclinical trials for patients and providers who are not physically located at the centerwherethetrialisbeingconducted.[127–130]Althoughmanyof safetynet. thesestrategieshavebeenconsideredfordecades,theCOVID-19 pandemicforcedtheirrapidadoption.Notably,theiradoptionearly inthepandemicwasassociatedwitharapidreboundintreatment

POTENTIALSOLUTIONS

Takentogether,issuesofremotegeography,limitedavailabilityof **cancer care centers, the need to BM quantum computers leverage furniture long distances, lower income,** andgreaterpovertycreatespecialchallengesforpatientswithcancerfromruralareasastheypertaintobothtrialparticipationand receiptofquality,nontrialoncologiccare.Theintersectionofthese domainscontributestoasystemicstructuraldifferencebetween ruralandurbancommunities,whichhasbeentermed*structuralurbanism*byProbstandcolleagues.[113]Thesestructuralbiasesresult fromamarket-basedapproachtohealthcarefunding;afocuson optimizingpopulation-levelhealthoutcomes,withanaccompanying emphasis on large population centers; and an inherent inefficiency of

trialenrollments,whichreturnedtonearpre-COVIDlevelsbythe endof2020withlittlelong-termconsequencesfortheintegrityof trialconductaccordingtofeedbackfromsponsors.[125,131]

The COVID-19 experience provides broader lessons for how the quality of cancer care might be improved—and made more accessible—forindividualsinruralareas.Thepotentialpromiseoftelemedicine washighlightedbytheInstituteofMedicinenearly30yearsago.[132] The use of telemedicine has had substantial study in both national and internationalsettings.[133–144]Modelsfortelemedicinevisitsinclude directpatientcontactwithorwithoutanaccompanyinginteraction with a local clinician or, alternatively, through the use of nonreal-time videocommunicationsordatatransmissions.[134,139]Patientshave commonlyexpressedsatisfactionwithtelemedicine,includingits delivering health care convenience, efficiency, communication, privacy, and com-

To improve cancer health care access and outcomes for patients withcancerfromruralareas,thehealthcaresystemmustbeori-entedtofindwaystobringcaretothepatientsor,alternatively,to improvewaystobringthepatientstocare,whilealsoprovidingpa-tientstheresources,includingthephysicianresources,toaccess care.Moreover,despitethechallenges,establishingaresearchcul-ture within rural communities can have multiple benefits that extend toaspectsofrurallivingbeyondthetreatmentofasingledisease, includingincreasedeconomicactivity,improvedhealthbehaviors, andimprovedoverallhealthparameters.[114–116]Thisorientation servesastheframeworkfortherecommendationstofollow(see

fort.[133,135,138] The effectiveness of telemedicine models in care de-liveryhasbeendemonstratedfordeliveringpsychiatricmedications, emergency care, and trauma care.[136,137,140]

In cancer, tele-oncology use was identified as superior to usual care for a range of patient quality-of-life outcomes, resulting in reduced depression and distress.[145] It has been demonstrated that tele-oncology models improve access to care and decrease health care costs.[139] The routine use of telemedicine could help change the landscape of oncology care access for patients with cancer from rural areas. However, its use is constrained in two fundamental ways. First, digital medicine and tele-oncology models are only as effective

Table 3). As the technology available at rural sites and in the homes of rural residents. Efforts to expand high-speed internet access through Congressional legislation have been a necessary adjunct to building

Digital communications and telemedicine

Modern digital and telecommunications tools could play a vital role in improving healthcare access for individuals from rural areas. Evi-dence from the conduct of clinical trials during the COVID-19 pandemic provides a guide. Early in the pandemic, participation in cancer clinical trials dropped dramatically.[117,118] Patients with a cancer diagnosis reported being fearful of coming to clinics while the pandemic raged.[119] At the same time, staff at institutions were occupied with COVID-19 cases, had themselves grown ill, or stayed home as a safety precaution.[120–122] In response, major federal

out capacity for telehealth models in rural areas.[146]For instance, the 2021InfrastructureInvestmentandJobsActincludedfundingto createtheBEAD(BroadbandEquity,Access,andDeployment)pro-gram,whichaimstoprovidefundingtostatestoexpandbroadband access.[147,148]Sucheffortswillneedtocontinue.Also,Medicare-fundedtelehealthvisitsincreased63-foldduringtheCOVID-19 pandemic,facilitatedbyCongressionalapprovalofflexibilities aroundMedicarecoverageoftelehealth.[149]Thesepandemic-era flexibilitiesareatriskofbeingdiscontinued.[150]Congressshould aimtocreateconsistentreimbursementmodelsfortelehealthcon-sultationsgoingforward.

TABLE3 Potentialsolutionsandassociatedpolicyframeworkstoadvancethereceiptofqualityoncologiccareinruralpopulations.

Issue	Potentialsolutions/strategies	Federal/
Digitalcommunicationsand telemedicine	Expandhigh	Congressionallegisl Access,andDeploy
	Telehealth	Medicarecoverageo
Hospitalaccess	Federalsupportforruralhospitals	Critical
Regionalizationofcare	Facilitaterelationshipswithlargenetworks	Partnershipswithth
	CommunityOncologyResearchProgram	
	Ruraloncologyhomemodels	
	Carecoordinationandhub-and-spokemodels	EnhancedreimbursementthroughMedicareorother programs
Physiciantimeandpractice expenses	Improved reimbursement for time and practice expenses	Increase Medicare and Medicaid geographic practice cost indicesforruralproviders
Drugpricing	Subsidizeddrugpricing	Reformthe340BDrugPricingProgram
		Assignthe340BDrugPricingProgramtoindividual patientswhomeetindigencycriteria,ratherthantoan institution
Specialtycare	Travelermodel	
Physiciancompensationand reimbursement	Loan-forgivenessprograms	ExpandtheNationalHealthServiceCor CommunityLoanRepaymentProgramt oncology
	Providefacilitiesatsubsidizedrates	
Emergencycare	Ensuretheavailabilityofemergencycare	CA hospitals; for non-CA hospitals, identify a to facilitate the support of emergency care (i.e., oncology medicalhomemodel)
Biomarkertestingandnext-generationsequencing	Enhanceduseofdigitalcommunicationstofacilitate genomic tumor board interactions using a hub-and-spoke modeltolargerinstitutions	
	Expandeddatacollectiontoenhanceidentificationof treatmentoptions	NationalLungCancerRoundtablemodel(expandedto includepatienthistories)
	Improvedaccesstocomprehensivetumorprofiling	Createlarge-scalenetworkforcomprehensivetumor profiling
Clinicaltrialparticipation	Promotetheconductofclinicaltrialsincommunity, underserved,andruralsites	NCI'sCommunityOncologyResearchProgram
	Applicationofdecentralizedclinicaltrialelements	USFoodandDrugAdministrationGuidance( )
	Streamlinetrialeligibility	Nationalworkinggroups(AmericanSocietyofClinical Oncology/Friends of Cancer Research/US Food and Drug Administrationjointtaskforce)
	Patienteducation	National,advocacy,andinstitutionalresourcesand programs
	Patientreimbursementtominimizeoreliminatetrial directandindirectcostsasabarriertoparticipation	

Criticalaccesshospitals

Criticalaccesshospitals(CAHs)offeravaluableresourcefor addressingtheneedsofcancerpatientslivinginruralareas.[151]The Centers for Medicare and Medicaid Services (CMS) designates CAHs

ashospitalslocatedinruralareasmorethan35milesfromthe nearesthospitalormorethan15milesinareaswithmountainous terrain or only secondary roads.[152]They must maintain no more than 25 inpatient beds and an annual average length of stay of ≤96 hours perpatientforacuteinpatientcare;furthermore,theymustfurnish

24-hour emergency care services 7 days a week. The requirement for emergency care services is a key element for oncology care in rural areas because the cost of maintaining an emergency department and the difficulty of finding staff can otherwise be substantial.

CAHs represent over one-half of all hospitals in rural communities.[153] However, they are underused compared with CMS-designated Prospective Payment System hospitals—which reimburse based on a predetermined, fixed amount—in both rural and urban areas.[154] By using data from the American Hospital Association Annual Survey, Hung and colleagues demonstrated that, in 2017, the availability of chemotherapy services was notably lower among CAHs (27.5%) compared with rural (46.5%) or urban (64.5%) Prospective Payment System hospitals.[153] Access to radiation services among CAHs was even lower (3.6% vs. 30.6% vs. 48.4%, respectively). Moreover, trends for both modalities were largely unchanged compared with the prior decade (2008). These findings highlight the limitations in specialized discipline and technical training required to provide oncology services in rural areas, a pattern that has persisted

facilitatetrialparticipation.AnexampleistheNewMexicoCancer CareAlliance,whichprovidesoncologycareinunderservedrural areas of New Mexico. The University of New Mexico Comprehensive CancerCenter,anNCORPmember,servesasthecentraltrialhub, andnineaffiliatesitesthroughoutthestatearethespokes,withan emphasisonlocalcustomsandlocalneeds, whichareimportantina statewithalargeminoritypopulationincludingapproximately40% Hispanicand10%NativeAmerican.[49]Similarly,amodelthatuses satelliteclinicstomainpracticesmaybeaneffectivewaytoextend trialparticipationtomoreruralareas.OneexampleistheONCare Alliance,acollaborativegroupof33independent,physician-owned practicesthroughouttheUnitedStatesthatserveslocalcommunity patients.[160]Because these models may only apply to sites within the samehealthcaresystemorpractices,adaptingthesemodelsfor cross-systemusecouldenhancetheirutility.[161]

Thehub-and-spokemodelorothermodelsforsharedregional-ization of care could be better supported through reimbursement for carecoordinationthroughfutureiterationsoftheCMSOncology overtime. CareModel,suchasitssuccessorprogram,theEnhancingOncology Model.[162,163]Theseprogramsincentivizeproviderstoimprovecan-cercarecoordination,ensureappropriatecare,enhancethepatient

Theregionalizationofcare

TheabilityofCAHstoprovideoncologycareservicescouldbe augmentedbyrelationshipswithotherproviderinstitutionsorby partneringwithoncologypracticestoprovideservices,suggestinga roleforimprovedregionalizationofoncologycarethroughshared carecoordination.[153,155]Onemodelcouldbepartnershipswith participating institutions in the NCI's Community Oncology Research Program (NCORP), which provides dedicated outreach to community institutions.[156]This program also provides access to clinical trials for

experience,andsupportdecisionmakingamongprovidersandcare settings.[164,165]These programs emphasize care coordination through physician–providergroupsunderthetraditionalMedicarefee-for-serviceprovidermodel.Althoughsomecoordinationservicesare reimbursablethroughMedicarebillingclaims,CAHscouldnotbe reimbursedundertheconstraintsoftheMedicarepaymentsys-tem.[153,166]In the absence of this, payment mechanisms to encourage carecoordinationsupportforunderservedruralhospitalsand physicianpracticegroupscouldservetopromulgatemultipleave-nuesofsharedcaremodelsacrossinstitutionsandclinicianswithin

patientswithcancerinruralareas.Forinstance,theUniversityof regions.
KansasCancerCenterhaspartneredwiththeMidwestCancerAlli-
ance to form a designated rural NCORP site, with the specific mission
toincreaseparticipationinclinicaltrialsforruralpopulationsin
Kansas.[157]ForoneoftheNCI'sclinicaltrialnetworks,fully19%of
patientsenrolledinclinicaltreatmenttrialsovernearly4decades

were from rural areas, demonstrating the reach of this trial program—as well as its predecessor program, the Community Clinical Oncology Program—outsideurbancenters.[34]Moregenerally,theincreased applicationofdigitaltelemedicineandcommunicationsstrategies couldallowinteroperablerelationshipsthatwouldbetterfacilitate combined services for rural patients with cancer. A recently proposed, community-based*ruraloncologyhome*modelwouldaimtoleverage advanced

Federalsupportforcliniciantimeandpractice expenses

-practiceprofessionals,carecoordination,anddigitaltele-health technologies to enhance the accessibility of specialized cancer care for rural patients.[158] Anotherstrategyisthehub-and-spokemodel,whichhas particular application to rural areas. Under a hub-and-spoke model, a centralhubfacilitatestheutilizationofmorecomplexcareservices nototherwiseavailableatspokesites,withpatientsatspokesites receivingtheirbasiccareserviceslocallyandthendirectedtothe

Rural oncology practices with low patient volumes are barely able to maintainfinancialviability,withamedianoverallprofitmarginof 0.1%in2017.[167]Currently,MedicareandMedicaidgeographic practicecostindicesbasemedicalreimbursement(forcliniciantime andeffortandpracticeexpenses)inpartoncost-of-livingfactors, suchasapartmentrentinthearea.[168,169]Thishasresultedinlower prices paid for medical services in underserved or rural communities. Thispatternisfurtherexacerbatedbythehigherprevalenceofpa-tientswithMedicareandMedicaidinruralareas.Yet,evenas reimbursementisofteninadequate,therelativecosttoclinical practicesofprovidingcomprehensiveoncologycareisroutinely higherinruralcommunities,partlybecausetherearenoeconomies ofscale.Forinstance,thecapacitytoprovideandmaintainthe equipment required for radiation services depends on the number of patientsservedbyaclinicandcanbecost-prohibitiveifthevolume

centralhubforfullercare.[159]Ahub-and-spokemodelcanalso islow.

TheMedicare and Medicaid geographic practice cost indices are anticipatedtobeupdatedin2026.[168]Werecommendrecalibrating theindicestobetterreflectthefinancialchallengesofproviding oncologiccareinruralareas.

onetothreeoutreachvisitspermonth)orhigh(greaterthanthree outreach visits per month) frequency.[175]For patients, this resulted in anaveragereductionintraveltimeforchemotherapyandradio-therapyservicesofabout15minutespertrip.

The traveler model can facilitate trial participation. The Missouri BaptistMedicalCenterfacilitatesimprovedruralparticipationin

Drugpricing

trialsbysendingmedicaloncologiststoruralsitestosupport participationintrials.Theoncologistsprovidetrainingandsupport

Drugpricingisacommonlycitedburdenforphysicianswhoprovide oncologyservices.[170]ThePublicHealthServiceActincludesapro-visioninSection340Bmandatingthatdrugmanufacturerspartici-pating in Medicaid must sell outpatient drugs at discounted prices to specificallydesignatedhealthcareorganizationsthatservemany uninsuredandlow-incomepatients.[171]Thisprovisionisespecially relevant to the needs of rural patients with cancer because it includes CAHsandruralreferralcenters.Thisprogramhasbeeneffectivein promotingoncologyprogramsinruralareas.Onestudyindicated that rural hospitals newly enrolling in the 340B program were about twice as likely (17% vs. 9%, an absolute 8% increase) to add oncology services,althoughthiseffectpredominantlyoccurredinstatesthat adoptedtheACAMedicaidexpansionorhadloweruninsurance rates.[172]As noted by Owsley and Bradley, the more limited increase

forlocalproviders.Inpartbecauseoftheseefforts,fully36%of consultingclinicsmodelinIowaaimstostafflocalruralclinicswith chemotherapymorethandoubled.Internationally,asdemonstrated primarycareandruralhospitalsettingsinAustraliaindicatedthe servicescanenhanceotherelementsofthecancercarecontinuum, suchascancerscreening.TheOhioStateUniversityComprehensive Cancer Center coordinates a mobile mammography program to bring mammographyaccesstowomenlivinginruralareasofAppalachia. Theprogram,whichhasscreenedthousandsofwomen,includes screening and navigationservices for women withabnormal findings inoncologyservicesinsocioeconomicallyvulnerableareassuggests onscreeningresults.[49] thatotherprogramscouldhelpsupportoncologyserviceadop- tion.[172]Anotherpossibilitymightbetoassign340Bpricingtoindi- vidualpatientswhomeetindigencycriteriaratherthantoan institution.Thiswouldallowanyoncologypractice,whetherornot hospital -based,tobetterserveindigentpatients.Removingthe ability of hospitals to use 340B drugs for fully insured patients would enhancetheoriginalintentofthelawandcurbtheacquisitionof oncologypracticesbyhospitals.

Specialtycare:Thetravelermodel

Foroncologyclinicians,creatingasustainablepracticeinarural setting can be both financially prohibitive and personally challenging. Medicalprofessionalsoftenemergefromschoolwithlargeamounts ofpersonaldebt,whichcanexacerbatephysicianburnout.[173]Physi- patientsenrolledintheirtrialsarefromruralareas.[49,176]Avisiting specialtyoncologyclinicians;underthismodel,localaccessto inasystematicreview,researchintospecialistoutreachclinicsin potentialtoimproveaccessandevenoutcomes.[177]Mobilecare ciansandadvancedpracticeprovidersmaybepersonallymotivated toliveinareas—especiallyurbanareas—whereraisingafamilyis mostconvenientandcanincludeopportunitiesfortheirspousesto findjobsandtheirchildrentoattendadequatelyresourcedschool systems. Despite efforts to recruit physicians to rural areas with large sign -onbonuses,medicalschooldebtforgiveness,andotherin- centives,theresultingmedicalcommunityisoftenunstable,lacking permanentoncologyspecialists.[48,170,174] Oneapproachtoaddressthisisatravelermodelforoncology specialtycare.Physiciansworkinginunderservedareasmaybeable to stay for a few days while other staff remain local. In a recent study examiningtheoutreachofoncologyprofessionalsinruralareas, aboutoneinfouroncologistswerecharacterizedashavingtraveled to provide care, nearly one half of whom traveled with medium (from

Physiciancompensationandreimbursement

Giventhesparseandtransitionalnatureoftheruraloncology workforce,otherapproachestocompensationandreimbursement maybeimportanttoexplore.Loan-forgivenessprograms,whichare commonforphysicianslocatingtoruralareas,couldbeboth increasedandindexedtotimespentintherurallocation.[178,179,180] Onepossibilitywouldbetoexpanddisciplineseligibleforthe Na-tionalHealthServiceCorpsRuralCommunityLoanRepayment Program to also include oncology. Communities in need of physicians couldprovidethebuildingsandpropertyforofficespaceatsubsi-dizedratesoronlease-to-owntermsifthephysicianstaysfora minimum number of years. Removing the overhead cost of a practice facility could make it more attractive to live in a rural or underserved area.Owningpropertyinanareacouldprovidetheeconomic incentiveforaphysicianorgroupofphysicianstocommittoadu-rablestay.Theestablishmentofafinanciallyinvestedrelationship withacommunitycouldfosteravirtuouscycleofinvestment.With theadventofastablemedicalcommunity,demandforhospitalser-vices,suchaslaboratoryorimagingservices,willincrease,further stabilizinghospitalfinances.Mechanismstopromotesuchmeasures arecurrentlyunknownandwoulddependonpolicymakersto explore.Ifanyofthesestrategiesareundertaken,ineithersmall, pilot tests or larger scale efforts, mechanisms for program evaluation should be incorporated into their design to collect data and evaluate theperformanceofthesemodels.Evaluationsshouldaimtounder-stand whether, and to what extent, outcomes are improved, including theirpotentialforimprovingclinicaltrialenrollment.

Emergency care Society's Project ECHO model.[192] Such models could be extended to also include relevant data on patients' clinical histories to expand

CAHs are required to provide emergency services 24/7. This requirement is critical for providing oncology care in rural areas because the cost of maintaining an emergency department and the difficulty of finding staff can otherwise be substantial. For hospitals that are not designated as CAHs, however, a mechanism to support emergency care should be identified because emergency department services can reasonably be viewed as services contributing to the common good, such as fire or police departments, which are other-wise supported by local tax dollars. An oncology medical home model, in theory, may minimize the number of emergency department visits, hospital admissions, and readmissions.[181–183] Providing the infra-structure and payments to a rural practice for these additional ser-vices may be a cost-effective way to improve care without training

possibletreatmentoptions.

AkeyrecommendationoftheCancerMoonshotBlueRibbon Reportwastheideaofestablishingalarge-scalenetworktooffer patientscomprehensivetumorprofiling.[3]Thisproposedrepository wouldenablepatientstomorereadilyparticipateinclinicaltrialsto accessthenewestavailabletreatmentsbasedontheirgenomic profileandimmunotype.Thisideawasmotivated,inpart,by concerns about known disparities in access to comprehensive cancer testingandnoveltreatments,especiallyforthosewithlimited socioeconomicmeansandruralpopulations.[184–188]Theemphasisof the Blue Ribbon Report was on participation in clinical trials because thepenetrationofgenome-drivenoncologyinpracticeatthetime waslimited.[193]However,biomarker-directedtherapyhasrapidly

localresources. expanded in routine practice. Thus a registry of this kind could also be used to support cancer care options outside of trials and would have a particular benefit for underserved rural communities.

Biomarkertestingandnext-generationsequencing

Numerousstudieshavedemonstratedthattheuseofbiomarker testingandnext -generationsequencingtodeterminetreatmentfor patientswithcancerislesscommoninunderservedareas.[184–188]A systematic review and meta-analysis indicated that patients with low socioeconomicstatuswere14%lesslikelytohavepredictive biomarker tests and 17% less likely to undergo biologic and precision

Trialparticipation:Accessandeligibility

Tohelpovercomethetyrannyoftraveldistanceforpatientswith cancerfromruralareasandtofindwaystobetterbringpatientsto trials,sponsorsshouldmoreroutinelyprovidethefundstotravelto participateinatrial,includingthecostsrelatedtohousing,trans-

therapy.[188] These disparitiesareimportant because carede- portation,parking,and

terminations,inandoutsideoftrials,areincreasinglybasedonthe genomicsofthepatient'scancer,includingastheypertainto receivingimmune -targetedormolecular-targetedtherapies.

Physicianfactorscouldplayanimportantroleinreducedutili-zationamongruralpatientswithcancerbecauseoncologistsare oftenchallengedtomaintaintheday-to-dayneedsofthepatients theyserve,letalonekeepingapprisedoftherapidlyevolving biomarker testing landscape that can point to a specific treatment for somecancers.Digitalcommunicationsmayofferonesolution.A genomictumorboardconsistsofcancergenomicexpertswhoassist oncologistsininterpretingtestresultsandidentifyingpotential therapeuticstrategiesbasedonactionabletargets.Itiswidely regardedasaneffectiveeducationalanddecision-supporttooland, withtheaidofteleconferencing,couldbeofparticularrelevanceto rural clinicians seeking treatment guidance.[189–191]Such a mechanism wouldbeespeciallyusefulforsitesalreadylinked(say,thorougha hub-and -spokemodel)tolargerinstitutions.Theseinteractiveses-sionswithcolleaguescouldsupportruralpracticestochoose appropriatetreatmentsbasedonpertinentgenomicdataandthe mostup-to -dateinformationinguidelines(suchastheNCCN). Artificialintelligencesoftwareembeddedinteleconferencingtools couldbeusedtosummarizediscussionsfordocumentationafter review and confirmation by providers. The American Cancer Society, throughtheNationalLungCancerRoundtable,hasconductedase-riesofstate -wideprojectstoexpandclinicians'knowledgeabout biomarkertestingforlungcancerusingtheAmericanCancer

soredtrials,whichrepresentalargemajorityofclinicaltreatment trials,providetravelsupportforpatientsinsomeinstances.[33]Offeringfinancialassistancetopatientswhowishtoparticipatein clinicaltrialsisacrucialstrategyforenhancingthegeographicdiversityoftrialcohorts,oneoftheaimsoftheFDA'sguidanceto industry about improving diversity in clinical trials.[196]One limitation isthatspecializedprocedures,suchassurgery,mayneedtobe performedbeforeparticipatinginastudytomeeteligibilitycriteria. Becausetheseproceduresarenotincludedinthetrial'sspecified therapies, anytravelcostsincurredforreceivingpretrialprocedures wouldnotbecoveredbythetrialsponsors.

Alternatively,theincreasedconductoftrialsincommunitysites wouldsatisfytheimperativetobringtrialstothepatients.Suchefforts,asrepresentedbytheNCI'sNCORPprogram(especiallythe minority-underservedNCORPs),havebeensuccessfulingenerating representativeenrollmentofruralpatientsinfederallysponsored trials.Onestudy,representingover40yearsofenrollmentto federallysponsorednetworkgrouptrials,demonstratedthatthe same proportion of individuals were from rural areas compared with theruralpopulationintheUnitedStates.[34]

Theapplicationofdecentralizedclinicaltrialstrategieswillbe vitalforbringingtrialparticipationopportunitiestowherepatients live.Digitaltechnologywouldbringthepotentialtoimproveaccess to trials for all patients and thereby to conduct trials more rapidly in morediversesetsofpatients,includingthosefromruralareas. Federalagenciesandadvocacyorganizationshavecalledforthe

increaseduseofdecentralizedclinicaltrialaspectstoeasethe participationofpatientsandproviders.[197–201] Trialeligibilitycriteriaoftenexcludepatientswithahigher burdenofcomorbiddiseases.Therefore,improvingaccesstoclinical trialsforindividualsinruralareas,whotendtohaveahigherprev-alenceofnon-cancerhealthconditions,willpartlydependonefforts tomakethesetrialsmoreinclusiveandaccommodatingtopatients with diverse health needs. A joint effort among the American Society forClinicalOncology,theFriendsofCancerResearch,andtheFDA hasresultedinrecommendationstomodernizeeligibilitycriteria.[202]

distortperceptionsofrisksandjudgments.However,theFDAhas articulatedthepositionthatpayingresearchsubjectstoparticipate instudiesisanacceptablepracticeingeneral.[215]Severalmodels abouthowtoreimbursepatientsfortrialparticipationthatwould alleviate concerns about undue influence have been proposed. These include a reimbursement model, wherein patients are reimbursed for theiractualexpenses;andawage-paymentmodel,whichaimsto financially compensate patients for their time, effort, discomfort, and studycontributions.[216]Underanypaymentmechanism,theultimate aimistoremoveancillarycostsasafactorinpatientdecision

The aim, in two cycles of effort, was to expand clinical trial access to making.[194]

individuals previously excluded from trials, including those with brain Despitethechallengesofconductingtrialsinruralareas,the metastases, human immunodeficiency virus infection, organ establishmentofaresearchculturewithinruralcommunitiesmay

dysfunction,priormalignancy,andpoorperformancestatus,among several other criteria.[202–207]The recommendations have changed the waytrialeligibilitycriteriaaredetermined;onestudyindicatedthat trialeligibilitycriteriaarenowmoreinclusiveofpatientswithbrain metastases.[208]Ongoing efforts to design more inclusive clinical trials willenhanceaccessforruralpatientswhohavecancer.This improvementwillalsoincreaseconfidencethattrialresultsare relevanttobothruralandurbanpatientswithcancer.

Clinicaltrialdesignsshouldprioritizepatientconvenience.This

havebenefitsthatextendtoaspectsofrurallivingbeyondthe treatment of a single disease. These may include increased economic activity,improvedhealthbehaviors,andimprovedoverallhealth parameters.[114–116]Viewedinthiscontext,theimprovedconductof clinicaltrialsinruralareasbringsfullcirclethepredicatethatthe conduct of cancer clinical trials is an integral element of the delivery ofqualityoncologiccaretoruralcommunities.

includesprovidingflexiblefollow-upschedulestoaccommodatethe demandsofdailylifeandworkinruralareas,wheretransportation

CONCLUSION

options may be limited and taking time off work can be more difficult. Given a culture of mistrust of medical science, attempts to demystify participationinclinicaltrialsarecrucial.[209]Thiscanbeachieved through educational efforts aimed at informing health care providers and their patients as well as the families and friends of potential trial participants.[210]Proactiveeducationaleffortstohelpilluminatethe nature of clinical trials could yield outsized benefits for patients who may consider trial participation a good option for their treatment and couldhelpovercomemistrust,aswellaslackofknowledge,about clinicaltrialparticipation.Inpart,thisisrelevantbecausemanypatientsaresimplynotawarethatatrialmaybeanoptionfortheir care,withonestudyindicatingthat80%ofpatientswereunaware thataclinicaltrialmightbeapossibility.[29,211]Variousorganizations provideexpertiseand resourcesin thisarea,includinggovernmental agencieslikeetheNCI;advocacyorganizations,includingtheAmericanCancerSocietyandtheAmericanSocietyofClinicalOncology; largecancercenters;andnonprofitorganizations(seeTableS1). Beyond travel costs, both direct and indirect costs to patients of trial participationcanbesubstantial.Patientsfacingsocioeconomicchallengesarelesslikelytoparticipateintrialsthanthosewithoutsuch challenges,withmultiplestudiesdemonstratingthatlowerincome patientswere30%lesslikelytoparticipateinclinicaltrialsthan higherincomepatients.[87,212,213,214]Thusclinicaltrialdesigners shouldminimizecostbyadoptingmorepragmatictrialdesignsand procedures,andsponsorsshouldhelpsupporttheeconomiccostto thepatientfortrialparticipation.Acommonreasonfornotreimbursingpatientsintrialsistopreventundueinfluenceontreatment decisions,asstatedintheUSCommonRuleforthe Protectionof

Thisreviewhighlightsissuesaroundaccessbothtocancerclinical trialsandtoqualityoncologiccareinruralareas.Ourpremisewas thatthebesttreatmentisreceivedinaclinicaltrial,andtheability toparticipateinclinicaltrialsalsoreflectstheabilitytoadoptnew clinicaltrialfindingsandtoprovidehigh-qualitycareinruralcom-munities.Ideally,clinicaltrialsshouldbeavailablewherepatients routinelyreceivetheironcologiccare.Providingaccesstoclinical trialsrequiresastrongmedicalecosystemandtheabilitytoprovide oncologiccarelocally.Effortstoresolvethesedisparitiesmust recognizethattheapplicationofefficiencyandscale—ahallmarkof marketforces—will,onitsown,beinadequatebecausethesame levelofoncologycareservicesisrequiredevenwhenpatientvol-umeislower.Nevertheless,theequitabledistributionofhealthcare betweenurbanandruralareasmaybringmutualbenefitsthat marketmetricscannotreadilyquantify.Ifruralpatientswithcancer canreceivecareclosetohome,theymayexperiencelessstressand greatertrust.Cancercareinformationmaybemoreefficiently distributed,improvingcommunityunderstandingofmedicalscience. Healthcaredollarswouldberetainedinruralcommunitiesrather thantransferredtourbancenters,improvingeconomicandhealth resilienceforcommunities.Participationinclinicaltrialscloseto homewouldspreadthemessagethatruralcitizenshavetheop-portunitytoreceivethesamecutting-edgetreatmentsastheirur-bancounterparts.Takentogether,thesefactorscouldcreatea virtuouscycleofmutualbenefittohelpbridgeruralandurban communities.Thus,providingqualityoncologycaretoruralpatients withcancerisnotjustanethicalandmoralimperativebutcould alsobringculturalandsociopoliticalbenefitsandshouldbeapri-HumanSubjects.Concernsexistthatfinancialincentivescould orityforpolicymakers.

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